Open problems:
Unphysicality of Hilbert spaces for quantum mechanics

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

From the brief

1. **Further problems.** The list of ways in which QSTO:H fails the base conditions is open-ended: "If the reader is aware of other problems that should be added, do contact us." (Brief, Sec. 1, footnote 1.)

2. **A physically adequate model.** The brief concludes that "the solution is not to find workarounds for these issues, but to fix the mathematical model to better reflect the physical requirements". A replacement for QSTO:H that satisfies [B]COBS, [B]WDQ and [B]STDIST is not given here; one candidate is the brief "Physical vector spaces for quantum mechanics". (Brief, Sec. 5.)

3. **Expectations of a general unbounded observable.** [B]COBS is about the expectation $\psi \mapsto \langle\psi|O|\psi\rangle$, while "boundedness is equivalent to continuity" is about everywhere-defined linear maps; an unbounded observable O is only densely defined. That the expectation of every unbounded observable is discontinuous is not proved. The formalization covers everywhere-defined linear maps (`not_continuous_of_unbounded`) and the oscillator example for N only. (Brief, Sec. 3, proof of Theorem 1, first paragraph; Log, "Results".)

From the formalization

1. [BLOCKS VERIFICATION: BRIEF ISSUE] **The conditions themselves.** [B]COBS, [B]WDQ, [B]STDIST and the inconsistency claimed in Theorem 1 are informal; only the mathematics behind each of the three arguments is checked. Precise readings are proposed in the last section; once they are confirmed, each part of Theorem 1 becomes a theorem about QSTO:H that can be checked in Lean. (Log, "Setting".)

2. **The unitary between the two frames.** The unitary U that implements the frame change $y(x) = \tan(\frac{\pi}{2}\operatorname{erf}(x))$ on wave functions, and the change of variables in the integral, are not formalized; only the densities $|\psi|^2$, $|\phi|^2$ and eq. (2) are. So is the general claim that "we can find a unitary U that connects operators with different domains". (Brief, Sec. 3; Log, "Differences".)

3. **Point-wise values.** That $\psi(x)$, and so $|\psi(x)|^2$, is not defined at each point of a ray of L^2 is not formalized. (Brief, Sec. 4; Log, "Differences", last item.)

4. **Commutator relations.** That $[X, P]/(i\hbar) = I$ cannot hold on the whole of $\mathcal{H}$ and must be restricted to a domain is not formalized. (Brief, Sec. 4; Log, "Differences", last item.)

5. **Domain of the Schrödinger equation.** That the Schrödinger equation holds only on the domain of H, since $P\psi$ and $V(x)\psi(x)$ need not be defined, is not formalized. (Brief, Sec. 4; Log, "Differences", last item.)

Proposed precise readings

The following readings are proposed to resolve the first item of the previous section. They are chosen so that the arguments of the brief go through as written, and they need to be confirmed against the intent of the brief.

Hypothesis. $\mathcal{H}$ is infinite-dimensional. QSTO:H allows any separable Hilbert space, but in finite dimension every self-adjoint operator is bounded and defined everywhere, so [B]COBS and [B]WDQ hold and Theorem 1 is false as stated.

Inconsistency. "QSTO:H is inconsistent with C" means: for every separable, infinite-dimensional complex Hilbert space $\mathcal{H}$, the model that QSTO:H builds on $\mathcal{H}$ fails condition C. For [B]STDIST, $\mathcal{H}$ is replaced by the family of state spaces of different systems.

[B]COBS.

- *States* are the nonzero vectors of $\mathcal{H}$, with the norm topology; this is equivalent to the quotient topology on $P(\mathcal{H})$, since everything below is invariant under $\psi \mapsto c\psi$.
- An *observable* is a self-adjoint operator O with domain $\mathcal{D}(O)$.
- The *expectation* of O is $\langle O\rangle_\psi = \langle\psi|O\psi\rangle \,/\, \langle\psi|\psi\rangle$, defined for $\psi \in \mathcal{D}(O) \setminus \{0\}$.

B COBS: for every observable O, $\psi \mapsto \langle O\rangle_\psi$ is continuous on $\mathcal{D}(O) \setminus \{0\}$ with the subspace topology.

- Claim to be proved: if O is densely defined, symmetric and unbounded (for every C there is $\psi \in \mathcal{D}(O)$ with $\|O\psi\| > C\|\psi\|$), then $\psi \mapsto \langle O\rangle_\psi$ is discontinuous at *every* point of $\mathcal{D}(O) \setminus \{0\}$; and every infinite-dimensional separable $\mathcal{H}$ carries an unbounded self-adjoint operator. The oscillator example of the brief is a special case. This also settles the third item of "From the brief".
- Alternative: define $\langle O\rangle_\psi$ on all of $\mathcal{H} \setminus \{0\}$ through the spectral measure of O, with values in $(-\infty, +\infty]$ for O bounded below. For the number operator this is $\sum_n n\,|c_n|^2 / \sum_n |c_n|^2$, which is lower semicontinuous but not continuous, so [B]COBS fails under this reading too.

[B]WDQ.

- A *frame* is a coordinate on the configuration space $\mathbb{R}$; a *change of frame* is a C^1 bijection $y = f(x)$ of $\mathbb{R}$ with $f' > 0$.
- A change of frame acts on states by the unitary $U_f \colon L^2(\mathbb{R}, dx) \to L^2(\mathbb{R}, dy)$, $(U_f\psi)(y) = \psi(f^{-1}(y))\,((f^{-1})'(y))^{1/2}$, so that $|\psi|^2$ transforms as a density (eq. (2)).
- The *corresponding quantity* in the new frame is the quantity of the same kind in the new coordinate: each observer measures position with the multiplication operator by their own coordinate, on its maximal domain. So X and Y are the same operator on $L^2(\mathbb{R})$, multiplication by the variable, applied to ψ and to $U_f\psi$. It is *not* $U_f X U_f^{-1}$: that operator has domain $U_f\,\mathcal{D}(X)$, and with it [B]WDQ would hold automatically.
- A quantity Q is *well defined* in a state ψ if $\psi \in \mathcal{D}(Q)$.

B WDQ: for every change of frame f and every state ψ, $\psi \in \mathcal{D}(X)$ implies $U_f\psi \in \mathcal{D}(X)$.

- Claim to be proved: for $f(x) = \tan(\frac{\pi}{2}\operatorname{erf}(x))$ and $\psi = (e^{-x^2}/\sqrt{\pi})^{1/2}$, $\psi \in \mathcal{D}(X)$ but $U_f\psi = (1/(\pi(1+y^2)))^{1/2} \notin \mathcal{D}(X)$. This also settles the second item of "From the formalization".
- Abstract form, for the sentence "we can find a unitary U that connects operators with different domains": if frames may be related by any unitary U, [B]WDQ says $U\,\mathcal{D}(O) \subseteq \mathcal{D}(O)$ for every observable O and every unitary U. By Hellinger–Toeplitz $\mathcal{D}(O) \neq \mathcal{H}$ for unbounded O, and a unitary that maps a unit vector of $\mathcal{D}(O)$ to one outside it violates this.

[B]STDIST.

- The *systems* are those with $n \geq 1$ degrees of freedom. For $n = 0$, $L^2(\mathbb{R}^0) = \mathbb{C}$ is finite-dimensional and the argument does not apply.
- The *mathematical representation* of a system is its state space $\mathcal{H}_n = L^2(\mathbb{R}^n)$, together with what QSTO:H builds from it alone: $P(\mathcal{H}_n)$, $D(\mathcal{H}_n)$, the self-adjoint operators, the Born rule and the von Neumann entropy.
- Two representations *tell systems apart* unless they are unitarily equivalent. Every part of the QSTO:H structure is carried along by a unitary $\mathcal{H}_n \to \mathcal{H}_m$, so unitary equivalence of the Hilbert spaces suffices.

B STDIST: for $n \neq m$ there is no unitary $\mathcal{H}_n \to \mathcal{H}_m$.

- Claim to be proved: for all $n, m \geq 1$ there is one (`L2_euclidean_iso`, already checked).
- Caveat for the brief: if the representation also includes the positions $X_1, \ldots, X_n$ and the momenta $P_1, \ldots, P_n$, then n is part of the data and [B]STDIST holds trivially. The argument needs the observables to be left out, and the brief should say so. Under the proposed reading, [B]STDIST fails even in finite dimension: the spin of an electron and the polarization of a photon are both represented on $\mathbb{C}^2$.